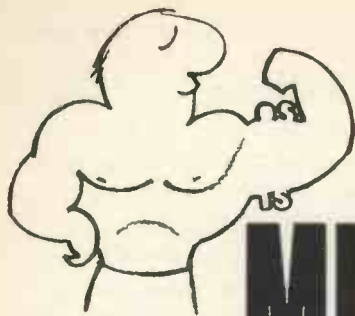




A MIGHTY MICROMITE

Tim Moore

The 77-68, a simple start to a large system

Introduction

Many people in this country have a considerable interest in, and a great desire to learn about and experiment with microprocessors. Unlike in the USA there is comparatively little spare spending money available for such luxuries, so there was a need for a low cost microprocessor system which could be built up cheaply and gradually into a complete and worthwhile system. The vast variety of microprocessors available cause a great deal of confusion to the newcomer to the field, but a few critical questions can rapidly clear this situation:—

- 1) Will it still be manufactured in five years' time?
- 2) Can it easily support a 'Basic' interpreter?
- 3) Is Software (Assemblers, Basic interpreters etc) readily available in the UK?
- 4) Is the price affordable?

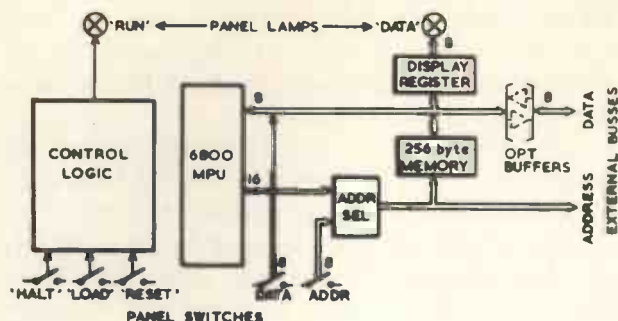
This immediately narrows the field down to the Z-80/8080 and the 6800. Of the two families the 6800 is simpler to begin with and better supported with software in the UK. The Z-80 is a very powerful but also very sophisticated and complex microprocessor, more suited to the experienced than the beginner.

The Simplest System

To make a 6800 microprocessor work, various additional items are needed eg:

- 1) Memory to store the program (which is to be executed).
- 2) A method of loading the memory and inputting data.
- 3) A method of outputting data.
- 4) All the necessary timing signals for the 6800.

These requirements can be satisfied with 256 bytes of memory, toggle switches, LED's, and a handful of TTL logic.



THE SIMPLEST SYSTEM

Operating Instructions:

Memory Change

Loading the memory with a program is accomplished by setting the 'Halt' switch which effectively turns off the 6800 and isolates it from the data and address bus. The 6800 achieves this by being able to make its data bus interface go high impedance. Also setting the halt switch causes X8 and X9 to select the address switches rather than the 6800 as the address inputs to the memory IC's X17 and X18.

Pushing the 'load' switch then causes whatever information that has been set into the data switches to be loaded into memory.

The contents of the memory can be inspected by setting up the address switches and observing the LED's.

Input/Output

The microprocessor, when it is executing a program, first of all selects a memory location and then either reads that byte of program or writes information to it. If location 'FF' hexadecimal (1111 1111) is selected and information read, the data switches are selected and whatever information they contain loaded into the 6800. Similarly if the 6800 writes into location 'FF' then the information is displayed via the LED's.

Running a Program

A 'Reset' causes the 6800 to jump to memory location 'FF' and there find the starting address of the program it is about to execute. If it is in the 'Halt' state it will then wait to be released with its starting address safely held in one of its internal registers.

Thus if the following program is loaded.

Memory Location

00	7C	INC
01	FF	
02	FF	
03	20	BRA
04	FB	

and the data switches set to zero, the 'reset' button

pushed and the halt switch set to run, then the instruction 7C is acted upon by the 6800 which asks it to go to memory location (FF) FF (only FF is acted upon because of the limited amount of memory present), reads it, (eg the switches) and then after incrementing it by one, writes it back into the same location (eg the LED's). So whatever the data switches are set to will, in a slightly modified form, appear on the LED's. The next instruction causes the 6800 to branch back to the starting address and repeat the operation in an endless loop.

Improvements on the basic system

Toggle switches and LED's are fine for demonstrating the binary nature of a microcomputer but a little tedious when 256 bytes of program have to be entered. The first improvement is to remove the LED's and replace them by two seven-segment LED displays and thus read Hexadecimal code directly. The second improvement is to build a hexadecimal keyboard that automatically advances the address switches and depresses the 'load' switch.

Constructional details

The printed circuit board is 8" by 8" and has a 78-way gold plated 0.1" pitch edge connector. This size was chosen because it is supported by several UK manufacturers. This means that card frames, prototyping boards and edge connectors are readily available at a reasonable price. As there are only 24 IC's it is feasible to hand-wire this circuit together, if a suitable prototyping board is already available.

If a reasonably deep junk box is available to the constructor, the cost of this machine could be kept to under £50.00.

Back-up Support

Microprocessors are complex devices and do take quite a lot of understanding before anything can be done with them. To help with this problem a manual was written which describes 77-68 in far greater depth, the 6800 instruction set, and gives a range of software which can be used. A user group has been formed which is supported by a quarterly newsletter. This was considered vital because it enables users who live locally to help each other commission their machines and chase out any faults that may have been introduced during construction. In practice this has worked very well and the user group now holds all sorts of software and hardware extension information for distribution among its members.

The Expanded System

To be able to run a 'Basic' interpreter, it is necessary to attach a visual display unit or teletype-writer to the computer and to have at least 4 kilobytes of memory.

This is achieved by regarding the main board as simply the central processor board of a large system. All the major control lines of the 6800 have been brought out to the edge connector and both the data and address busses have been fully buffered. A 4 kilobyte RAM printed circuit board and a

'Soft' monitor board which interfaces the system to a VDU etc. Design information already exists for interfacing the 'Mikbug' monitoring ROM with a single step feature and various other additions are expected such as a single card VDU and a Dynamic RAM board.

The method of connecting these boards together is simplicity itself: attach the appropriate number of edge connectors to the card frame and then lay a piece of vero board across all the edge connector pins, connecting all the pin 1's together, and so on. This enables a very cheap, fast and reliable back plane to be constructed. This will be appreciated by anybody who has ever wired up a large backplane — it takes a long time.

Conclusions

77-68 is a 'live' project, every month brings news of another extension to this system. All the parts are readily obtainable and if any problems are encountered there is a choice of a hundred or so other users who can give impartial advice and help.

Summary

- * A 6800 based system for home construction.
- * Low cost approx. £50.00 with help from a junk box.
- * Can be fully expanded to a 'mainframe' micro-computer.
- * Fully supported with a user group.
- * Can be used with a simple switch interface or a Visual Display Unit.

(PCW We hope to feature another Mighty Micromite in the next issue of Personal Computer World. PCW)

77-68 Components List

Logic

3 off 7400	X4, X14, 24	1 off 7495	X2
1 off 7402	X19	2 off 74125	X22, X23
1 off 7404	X1	2 off 74157	X8, X9
1 off 7410	X3	*2 off 81LS97	X15, X16
2 off 7428	X5, X6	2 off 2112	X17, X18
1 off 7430	X7	1 off 6800	X13
2 off 7432	X11, X12		
1 off 7474	X10		
2 off 74LS75	X20, X21	* optional buffers	

Resistors

20 off 4.7k	$\frac{1}{2}W$
*2 off 1k	$\frac{1}{2}W$
14 off 430	$\frac{1}{2}W$
1 off 270	$\frac{1}{2}W$
2 off 220	$\frac{1}{2}W$
2 off 22	$\frac{1}{2}W$

Capacitors

3 off 33uF	6.3V tant
8 off 0.1uF	ceramic
1 off 33pF	

LED's 9 off

Switches

17 subminiature toggle switches
single pole 2 way S3-19
1 push single pole n.o. S1
1 push single pole c.o. S2

Crystal 5 MHz

* option

Low profile sockets 2 off 16 pin
1 off 40 pin

3 feet 22 s.w.g. wire and sleeving

1 foot 8 way ribbon cable

Printed Circuit Board

Edge Connector, 77 way + polarising key 0.1" single sided.

